HONGKANG LI

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Academic & Research Appointments

Postdoctoral Researcher, University of Pennsylvania Jun. 2025–Present
Supervisor: Prof. René Vidal

Postdoctoral Researcher, Rensselaer Polytechnic Institute Jan. 2025–May 2025
Supervisor: Prof. Meng Wang

Research Intern, IBM Research May 2023–Aug. 2023
Project: Theoretical Foundations of In-Context Learning and Chain-of-Thought
Supervisor: Dr. Songtao Lu, Dr. Hui Wan, Dr. Xiaodong Cui

Education

Rensselaer Polytechnic Institute, Troy, NY, United States Aug. 2019–Dec. 2024
Ph.D., Electrical, Computer, and Systems Engineering
Advisor: Prof. Meng Wang

University of Science and Technology of China, Hefei, China Aug. 2015–Jun. 2019
Bachelor of Engineering

Research Interests

Theory of Foundation Models: optimization, feature learning, generalization, and mechanisms of Transformers, state-space models, and generative models.
Efficient and Trustworthy AI: data- and inference-efficient learning, robustness, and interpretability.

Selected Honors & Awards

- **IEEE BigData Future Faculty Rising Star**, 2026
- **University Nominee (RPI), CGS/ProQuest Distinguished Dissertation Award**, 2026
- **IEEE Signal Processing Society Best Ph.D. Dissertation Award**, 2025
- **MLCommons Machine Learning and Systems Rising Star Award**, 2025
- ICML Gold Reviewer Award, 2026
- NeurIPS Top Reviewer Award, 2025
- ICLR Notable Reviewer Award, 2025
- **Founders Award of Excellence**, Rensselaer Polytechnic Institute, 2023
- **Belsky Award**, School of Engineering, Rensselaer Polytechnic Institute, 2023
- Zenghua Scholarship, Top 5% of USTC, 2018

Publications

** denotes equal contribution. Author name is bolded for readability.*

Lead / Co-Lead Author Publications

[**ICML'26**] **H. Li**, H. Min, R. Vidal, "Transformers Learn the Optimal DDPM Denoiser for Multi-Token GMMs."

[**ICML'26**] **H. Li**, S. Lu, X. Cui, P.-Y. Chen, M. Wang, "How Can Mamba Learn In Context with Outliers and Generalize Provably?"

[**AISTATS'26**] Y. Zhang*, **H. Li***, Y. Yao*, A. Chen, S. Zhang, P.-Y. Chen, M. Wang, S. Liu, "Visual Prompting Reimagined: The Power of Activation Prompts."

[**ICLR'25**] **H. Li**, Y. Zhang, S. Zhang, M. Wang, S. Liu, P.-Y. Chen, "When is Task Vector Provably Effective for Model Editing? A Generalization Analysis of Nonlinear Transformers." **Oral Presentation (1.8%)**.

[**ICLR'25**] **H. Li**, M. Wang, S. Lu, X. Cui, P.-Y. Chen, "Training Nonlinear Transformers for Chain-of-Thought Inference: A Theoretical Generalization Analysis."

[**ICML'24**] **H. Li**, M. Wang, S. Lu, X. Cui, P.-Y. Chen, "How Do Nonlinear Transformers Learn and Generalize in In-Context Learning?"

[**ICML'24**] **H. Li**, M. Wang, T. Ma, S. Liu, Z. Zhang, P.-Y. Chen, "What Improves the Generalization of Graph Transformer? A Theoretical Dive into Self-attention and Positional Encoding."

[**IEEE JSTSP'24**] **H. Li**, S. Zhang, M. Wang, Y. Zhang, P.-Y. Chen, S. Liu, "How Does Promoting the Minority Fraction Affect Generalization? A Theoretical Study of One-Hidden-Layer Neural Network on Group Imbalance."

[**SAM'24**] **H. Li**, M. Wang, S. Zhang, S. Liu, P.-Y. Chen, "Learning on Transformers Is Provable Low-Rank and Sparse: A One-Layer Analysis."

[**ICLR'23**] **H. Li**, M. Wang, S. Liu, P.-Y. Chen, "A Theoretical Understanding of Shallow Vision Transformers: Learning, Generalization, and Sample Complexity."

[**ICML'22**] **H. Li**, M. Wang, S. Liu, P.-Y. Chen, J. Xiong, "Generalization Guarantee of Training Graph Convolutional Networks with Graph Topology Sampling."

[**CISS'22**] **H. Li**, S. Zhang, M. Wang, "Learning and Generalization of One-Hidden-Layer Neural Networks, Going Beyond Standard Gaussian Data."

Collaborative Publications

[**ECCV'26**] T. Ding, Z. Xu, B. Haeffele, **H. Li**, R. Vidal, "Parsimonious Flow Matching for Efficient Image Generation."

[**ICLR'26**] M. Shandirasegaran, **H. Li**, S. Zhang, M. Wang, S. Zhang, "A Theoretical Analysis of Mamba's Training Dynamics: Filtering Relevant Features for Generalization in State Space Models."

[**NeurIPS'25**] J. Sun, S. Zhang, **H. Li**, M. Wang, "Contrastive Learning with Data Misalignment: Feature Purity, Training Dynamics and Theoretical Generalization Guarantees."

[**ICLR'25**] Y. Luo, **H. Li**, Q. Liu, L. Shi, X.-M. Wu, "Node Identifiers: Compact, Discrete Representations for Efficient Graph Learning."

[**TMLR'25**] J. Sun, **H. Li**, M. Wang, "Theoretical Learning Performance of Graph Networks: The Impact of Jumping Connections and Layer-wise Sparsification."

[**NeurIPS'24**] Y. Luo, **H. Li**, L. Shi, X. Wu, "Enhancing Graph Transformers with Hierarchical Distance Structural Encoding."

[**NeurIPS'23**] S. Zhang, **H. Li**, M. Wang, M. Liu, P.-Y. Chen, S. Lu, S. Liu, K. Murugesan, S. Chaudhury, "On the Convergence and Sample Complexity Analysis of Deep Q-Networks with ϵ-Greedy Exploration."

[**ICASSP'23**] H. Wan, **H. Li**, S. Lu, X. Cui, M. Danilevsky, "How Can Personalized Context Help? Exploring Joint Retrieval of Passage and Personalized Context."

Preprint

H. Li, T. Ding, U. Tadipatri, R. Vidal, "Finite-Sample Training, Generalization, and Memorization in Transformer DDPMs," under review

J. Sun, S. Zhang, **H. Li**, S. Liu, P.-Y. Chen, M. Wang, "Enhanced Generation in Diffusion Models via Joint High-to-Low Denoising Schedule and Model Sparsity with Theoretical Guarantees," under review.

A. Amini, **H. Li**, X. Cui, S. Lu, C. Ko, K. Murugesan, Y. Ma, M. Wang, "How Multi-Head Transformers Specialize for In-Context Learning: A Theoretical Analysis of Training Dynamics and Generalization," under review.

Research Mentoring

- **Tianjiao Ding**, Ph.D. student, University of Pennsylvania (advisor: René Vidal), Sep. 2025–Present. Coauthored [**ECCV'26**] "Parsimonious Flow Matching for Efficient Image Generation."
- **Mugunthan Shandirasegaran**, Ph.D. student, New Jersey Institute of Technology (advisor: Shuai Zhang), Sep. 2025–Dec. 2025. Coauthored [**ICLR'26**] "A Theoretical Analysis of Mamba's Training Dynamics: Filtering Relevant Features for Generalization in State Space Models."
- **Jiawei Sun**, Ph.D. student, Rensselaer Polytechnic Institute (advisor: Meng Wang), Jun. 2023–Present. Coauthored [**NeurIPS'25**] and [**TMLR'25**] publications listed above.

Invited Talks

- **International Conference on Next-Generation AI & Machine Learning**, Nov. 2026. "How Foundation Models Learn, Adapt, and Generate: A Mathematical Perspective"
- **City University of Hong Kong, Frontier of Artificial Network Seminar**, Jun. 2026. "Towards a Mathematical Theory of Foundation Models."
- **Illinois Tech ECE Graduate Research Seminar**, Oct. 2025. "Theoretical Perspectives of Efficient Learning for Large Foundation Models."
- **IBM-RPI FCRC Seminar Series**, Feb. 2025. "Theoretical Foundations of Chain-of-Thought with Trained Nonlinear Transformer Models."
- **NJIT Data Science Seminar**, Oct. 2024. "Theoretical Foundations of In-Context Learning and Chain-of-Thought with Trained Transformers."
- **IBM-RPI FCRC Seminar Series**, Feb. 2024. "Theoretical and Algorithmic Foundations of In-Context Learning Using Properly Trained Transformer Models."

Press Coverage

- **IBM Research Blog**, Jul. 2024. "IBM and RPI researchers demystify in-context learning in large language models."

Professional Service

- **Area Chair:** ICLR 2027; ICASSP 2026-2027; CPAL 2026-2027.
- **Conference Reviewer:** ICML 2022–2026; NeurIPS 2022–2026; ICLR 2024–2026; CVPR 2024–2026; AAAI 2025; AISTATS 2024–2026; ICASSP 2023–2026.
- **Journal Reviewer:** IEEE Transactions on Artificial Intelligence 2024; IEEE Transactions on Signal Processing 2024; Journal of Machine Learning Research 2026; TMLR 2023–2025.